

# Frontend Quiz

JCWD-3302

1. What is the main role of HTML in web development ?
  - a. **To define the structure and content of a web page**
  - b. To style the elements of a webpage
  - c. To execute JavaScript code
  - d. To store data in a database
2. Which HTML element is used to create a hyperlink ?
  - a. **<button>**
  - b. <link>
  - c. <a>
  - d. <img>
3. What does the flex-grow property do in CSS Flexbox ?
  - a. **Controls how much an element should grow to fill available space**
  - b. Defines the initial size of an element
  - c. Specifies if an element can shrink
  - d. Determines the main axis of the flex container
4. What is 'CSS Flexbox' and how is it used to layout elements on a web page?
  - a. **CSS Flexbox is a method used for form validation in CSS.**
  - b. CSS Flexbox is a method for arranging the layout elements in one dimension (row or column) flexibly
  - c. CSS Flexbox is a method for arranging background images within HTML elements.
  - d. **CSS Flexbox is a method for positioning element layout such as fixed, sticky, and absolute.**
5. What is the core concept of React that enables UI development using reusable pieces ?
  - a. **MVC Pattern**
  - b. **Component-Based Architecture**
  - c. Layout
  - d. **Object-Oriented Programming**
6. How do you define a functional component in React ?
  - a. **class MyComponent extends React.Component {}**
  - b. **const MyComponent = new React.Component();**
  - c. **function MyComponent() { return <> </> }**
  - d. **React.createComponent('div', 'Hello')**
7. What is meant by 'props drilling' in the context of React ?
  - a. **Props drilling is a way to organize prop data in a structured and efficient manner**
  - b. **Props drilling is a technique or method for passing props in React**
  - c. **Props drilling is the process of passing props data through multiple layers of component, which can make the code inefficient.**
  - d. **Props drilling is a method for passing props from child component to parent component in react, making the code efficient.**
8. What is the primary advantage of the Virtual DOM in React?
  - a. **It reduces direct updates to the real DOM**
  - b. **It increases the application's size**
  - c. **It makes components larger**
  - d. **It adds compatibility with HTML5**
9. How is useMemo used in React to optimize the performance of a component that fetches data from an external API ?
  - a. **useMemo is used to efficiently fetch data from the API every time the component re-renders.**
  - b. **useMemo is used to optimize performance by memoizing expensive operations or calculations that depend on values derived from API data.**
  - c. **useMemo cannot be used to optimize data fetching from an API.**
  - d. **useMemo is used to manage the rendering of React components.**

10. Which of the following is true about how state is updated in Zustand compared to Redux?
- Redux uses mutation, while Zustand uses immutable updates**
  - Zustand uses reducers like Redux for all updates**
  - Redux requires pure functions (reducers), while Zustand allows direct state modification via set**
  - Zustand uses dispatch to trigger actions like Redux
11. How do you access a React Context using useContext ?
- const {value, setValue} = useContext(MyContext);**
  - const [state, dispatch] = useContext();**
  - const context = useContextProvider();**
  - useContext(MyContext, state => state.value);**
12. How do you define a Server Component in Next.js App Router ?
- By using "use client" at the top of the file**
  - By omitting "use client" at the top of the file**
  - By exporting the component as ServerComponent**
  - By wrapping the component with useServer()**
13. Why should you avoid copying prop values into the component state?
- Because props values cannot be stored by the state in that component**
  - Because it would create two instance with the same state, which could lead to state desynchronization.**
  - Because the mutation process on state does not allow such mutation processes**
  - Because it would lead to duplicate instances with different state, while state synchronization remains intact but it consume a lot of memory.**
14. What is meant by 'Virtual DOM' in the context of react ?
- A javascript library used to facilitate creating virtual animation interactions in react.**
  - A temporary representation of the actual DOM structure used by React to improve rendering performance.**
  - A type of component in react used to replace structure in the form of functional components**
  - A DOM element used to render components in React.**
15. What is Server Side Rendering (SSR) in react, and what are its benefits?
- SSR is a method for managing global state in react.**
  - SSR is a method for rendering views on the server side before sending them to the client, which allows the application to load faster and have better SEO.**
  - SSR is a method for rendering views only on the client side before sending them to the server, which allows the application to load faster and have better SEO**
  - SSR is a method for rendering only animated views in react before sending them to the server, which allow the applications to load faster and have better SEO**
16. Which of the following statements is correct regarding page navigation in a Next JS project?
- Using Link from next/link is very good for SEO performance because page transition links can be detected by crawlers and have SPA behavior.**
  - There is no correct statement.**
  - Using Link from next/link is good for SEO performance because page transition links cannot be detected by crawlers.**
  - router.push is used for page navigation, which is very good for SEO performance because page transition links can be detected by crawlers and have SPA behavior**
17. Which method is used to fetch data on the server in a Server Component ?
- fetch() inside useEffect**
  - getServerSideProps**
  - fetch() directly inside the component**
  - useSWR**

18. Which URL structure is more SEO-friendly ?
- example.com/page?id=123&sort=asc**
  - example.com/?p=12345**
  - example.com/best-laptops-2024**
  - example.com/index.php?page=home**
19. Here is a functional component in react. How to complete this code with the correct way to manage state and display the state value when the button is pressed.

```
import React, { useState } from "react";
function Counter() {

  return (
    <div>
      <p>Count value</p>
      <button>Increment 1</button>{" "}
    </div>
  );
}
export default Counter;
```

- Use useState to initialize the count state and setCount, and use setCount to update the state when the button is pressed**
  - State is not required in functional component**
  - Use state to initialize the count state and setState, and use setState to update the state when the button is pressed.**
  - Use this.state to initialize the count state and this.setState to update the state when the button is pressed**
20. You want to access DOM elements in your react component. How would you use the useRef hook to achieve this?

```
import React, { useRef } from 'react';

function ElementManipulation() {
  const elementRef = useRef<HTMLDivElement | null>(null);

  const handleClick = () => {
    // complete code
  };

  return (
    <div>
      <button onClick={handleClick}>Ubah Warna</button>
      <div ref={elementRef} style={{ width: '188px', height: '108px', backgroundColor: 'red' }}></div>
    </div>
  );
}

export default ElementManipulation;
```

- You can use elementRef.style.color = 'blue' in the handleClick function to change the background color of the div element.**

- b. You can use `elementRef.current.color = 'blue'` in the `handleClick` function to change the background color of the div element.
- c. You can use `elementRef.style.backgroundColor = 'blue'` in the `handleClick` function to change the background color of the div element
- d. You can use `elementRef.current.style.backgroundColor = 'blue'` in the `handleClick` function to change the background color of the div element.

21. You have a functional component in react that has several hooks

```
import React, { useState, useEffect } from 'react';

function DataFetching() {
  const [data, setData] = useState<any[]>([]);
  const [loading, setLoading] = useState<boolean>(true);

  useEffect(() => {
    // complete code
  }, []);

  return (
    <div>
      <h1>Data yang diambil:</h1> {/* Corrected heading tag */}
      {loading ? (
        <p>Loading...</p>
      ) : (
        <ul>
          {data.map((item) => (
            <li key={item.id}>{item.title}</li>
          ))}
        </ul>
      )}
    </div>
  );
}

export default DataFetching;
```

Complete the following code:

- a. You can directly use data and loading to display data in the UI without needing to use `useEffect`
- b. In `useEffect`, you will send a HTTP request using `fetch` to fetch data from the server and populate data with the response. When the data is received, you will change the loading value to false
- c. You should add and create a new state inside `useEffect` to manage data and loading because you cannot use `useState` outside the return. After that, make an HTTP request directly inside `useEffect`.
- d. You should initialize data and loading with fake data from the start, and then update them when the actual data arrives.

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23. Take a look at the following code! The snippet below is from a Next.js version 14 project that uses the app router.

```
import { useState } from "react";

export default function Home() {
  const [data, setData] = useState<number>(0);

  return (
    <div>
      <button onClick={() => setData(data + 1)}></button>
      <div>{data}</div>
    </div>
  );
}
```

What is wrong with the code ?

- a. You need to define the page using class component .
- b. You need to define the page as client side using "use client" .
- c. The use of useState must be accompanied by useEffect in that component.
- d. The Function in the onClick must be wrapped in seperate function and initialized above before calling it.